

LAPORAN MODUL PRAKTIKUM PEMROGRAMAN SISI BACK END



Disusun oleh :

Nama : Irvan Cahya Nugraha
NIM : 245410034
Kelas : Informatika 1

**PROGRAM STUDI INFORMATIKA
PROGRAM SARJANA
FAKULTAS TEKNOLOGI INFORMASI
UNIVERSITAS TEKNOLOGI DIGITAL INDONESIA
YOGYAKARTA
2026 / 2027**

MODUL 7

SQL

A. TUJUAN PRAKTIKUM

1. Memahami perintah perintah dasar SQL untuk CRUD

B. PEMBAHASAN LISTING PRAKTIK

PRAKTIK

Praktik 1

```
C:\xampp\mysql\bin>mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 49
Server version: 10.4.32-MariaDB mariadb.org binary distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

```
Administrator: C:\WINDOWS\ × + ∨

MariaDB [(none)]> SHOW DATABASES;
+-----+
| Database |
+-----+
| datamahasiswa |
| information_schema |
| login |
| mysql |
| online_shop |
| performance_schema |
| phpmyadmin |
| produk |
| test |
| user_db |
+-----+
10 rows in set (0.002 sec)

MariaDB [(none)]> CREATE DATABASE storedb;
Query OK, 1 row affected (0.002 sec)

MariaDB [(none)]> SHOW DATABASES;
+-----+
| Database |
+-----+
| datamahasiswa |
| information_schema |
| login |
| mysql |
| online_shop |
| performance_schema |
| phpmyadmin |
| produk |
| storedb |
| test |
| user_db |
+-----+
11 rows in set (0.001 sec)

MariaDB [(none)]> DROP DATABASE storedb;
Query OK, 0 rows affected (0.037 sec)

MariaDB [(none)]> CREATE DATABASE IF NOT EXISTS storedb;
Query OK, 1 row affected (0.002 sec)

MariaDB [(none)]> DROP DATABASE IF EXISTS storedb;
Query OK, 0 rows affected (0.002 sec)

MariaDB [(none)]> CREATE DATABASE IF NOT EXISTS storedb;
```

```

MariaDB [(none)]> CREATE DATABASE IF NOT EXISTS storedb;
Query OK, 1 row affected (0.002 sec)

MariaDB [(none)]> SHOW CREATE DATABASE storedb \G
***** 1. row *****
      Database: storedb
Create Database: CREATE DATABASE `storedb` /*!40100 DEFAULT CHARACTER SET utf8mb4 COLLATE utf8mb4_general_ci */
1 row in set (0.001 sec)

MariaDB [(none)]> DROP DATABASE IF EXISTS storedb;
Query OK, 0 rows affected (0.003 sec)

MariaDB [(none)]> SHOW DATABASES;
+-----+
| Database |
+-----+
| datamahasiswa |
| information_schema |
| login |
| mysql |
| online_shop |
| performance_schema |
| phpmyadmin |
| produk |
| test |
| user_db |
+-----+
10 rows in set (0.001 sec)

MariaDB [(none)]> QUIT;
Bye

C:\xampp\mysql\bin>mysql -u root -p|

```

Pembahasan : Program praktik 1 membahas perintah dasar untuk membuat dan menghapus database. Perintah SHOW DATABASES digunakan untuk menampilkan seluruh database yang ada, CREATE DATABASE storedb digunakan untuk membuat database baru bernama storedb, dan DROP DATABASE storedb digunakan untuk menghapusnya. Penambahan IF NOT EXISTS pada CREATE dan IF EXISTS pada DROP berfungsi sebagai pengaman agar perintah tidak error jika database sudah ada atau belum ada. Perintah SHOW CREATE DATABASE storedb menampilkan detail perintah yang digunakan untuk membuat database tersebut.

Praktik 2

```

C:\xampp\mysql\bin>mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 50
Server version: 10.4.32-MariaDB mariadb.org binary distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> SHOW DATABASES;
+-----+
| Database |
+-----+
| datamahasiswa |
| information_schema |
| login |
| mysql |
| online_shop |
| performance_schema |
| phpmyadmin |
| produk |
| test |
| user_db |
+-----+
10 rows in set (0.002 sec)

MariaDB [(none)]> CREATE DATABASE `storedb` ;
Query OK, 1 row affected (0.015 sec)

```

```

MariaDB [mysql]> CREATE USER 'store'@'%' IDENTIFIED BY 'store';
Query OK, 0 rows affected (0.022 sec)

MariaDB [mysql]> GRANT ALL PRIVILEGES ON storedb.* TO 'store'@'%';
Query OK, 0 rows affected (0.002 sec)

MariaDB [mysql]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.016 sec)

```

```

MariaDB [mysql]> SHOW DATABASES;
+-----+
| Database |
+-----+
| datamahasiswa |
| information_schema |
| login |
| mysql |
| online_shop |
| performance_schema |
| phpmyadmin |
| produk |
| storedb |
| test |
| user_db |
+-----+
11 rows in set (0.002 sec)

```

Pembahasan : Program praktik 2 membahas pembuatan user database. Perintah CREATE USER 'store'@'%' IDENTIFIED BY 'store' membuat user baru bernama store dengan password store yang bisa mengakses dari host manapun karena menggunakan '%'. Perintah GRANT ALL PRIVILEGES ON storedb.* TO 'store'@'%' memberikan seluruh hak akses user store terhadap semua tabel di database storedb. Perintah FLUSH PRIVILEGES dijalankan untuk memastikan perubahan hak akses langsung diterapkan oleh server.

Praktik 3

```
MariaDB [mysql]> USE storedb;
Database changed
MariaDB [storedb]> SELECT DATABASE();
+-----+
| DATABASE() |
+-----+
| storedb    |
+-----+
1 row in set (0.006 sec)

MariaDB [storedb]> SHOW TABLES;
Empty set (0.001 sec)

MariaDB [storedb]> CREATE TABLE IF NOT EXISTS products (
  -> id INT UNSIGNED NOT NULL AUTO_INCREMENT,
  -> productCode CHAR(3) NOT NULL DEFAULT '',
  -> name VARCHAR(30) NOT NULL DEFAULT '',
  -> quantity INT UNSIGNED NOT NULL DEFAULT 0,
  -> price DECIMAL(7,2) NOT NULL DEFAULT 99999.99,
  -> PRIMARY KEY (id)
  -> );
Query OK, 0 rows affected (0.038 sec)

MariaDB [storedb]> SHOW TABLES;
+-----+
| Tables_in_storedb |
+-----+
| products          |
+-----+
1 row in set (0.001 sec)

MariaDB [storedb]> DESCRIBE products;
+-----+-----+-----+-----+-----+-----+
| Field      | Type                | Null | Key | Default | Extra           |
+-----+-----+-----+-----+-----+-----+
| id         | int(10) unsigned    | NO   | PRI | NULL     | auto_increment |
| productCode | char(3)              | NO   |     |          |                 |
| name       | varchar(30)         | NO   |     |          |                 |
| quantity   | int(10) unsigned    | NO   |     | 0        |                 |
| price      | decimal(7,2)        | NO   |     | 99999.99 |                 |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.019 sec)

MariaDB [storedb]> SHOW CREATE TABLE products;
```

Pembahasan : Program praktik 3 membahas pembuatan tabel products di dalam database storedb. Perintah USE storedb digunakan untuk berpindah ke database storedb, kemudian

CREATE TABLE IF NOT EXISTS products membuat tabel dengan lima kolom yaitu id bertipe INT UNSIGNED dengan AUTO_INCREMENT sebagai primary key, productCode bertipe CHAR(3), name bertipe VARCHAR(30), quantity bertipe INT UNSIGNED, dan price bertipe DECIMAL(7,2). Perintah DESCRIBE products digunakan untuk menampilkan struktur tabel yang baru dibuat.

Praktik 4

```
MariaDB [storedb]> INSERT INTO products VALUES (1001, 'PEN', 'Pen Red', 5000, 1.23)
;
Query OK, 1 row affected (0.016 sec)

MariaDB [storedb]> INSERT INTO products VALUES (NULL, 'PEN', 'Pen Blue', 8000, 1.25
), (NULL, 'PEN', 'Pen Black', 2000, 1.25);
Query OK, 2 rows affected (0.002 sec)
Records: 2 Duplicates: 0 Warnings: 0

MariaDB [storedb]> INSERT INTO products (productCode, name, quantity, price) VALUES
('PEC', 'Pencil 2B', 10000, 0.48), ('PEC', 'Pencil 2H', 8000, 0.49);
Query OK, 2 rows affected (0.002 sec)
Records: 2 Duplicates: 0 Warnings: 0

MariaDB [storedb]> INSERT INTO products (productCode, name) VALUES ('PEC', 'Pencil
HB');
Query OK, 1 row affected (0.002 sec)

MariaDB [storedb]> SELECT * FROM products;
+-----+-----+-----+-----+-----+
| id    | productCode | name      | quantity | price  |
+-----+-----+-----+-----+-----+
| 1001  | PEN         | Pen Red   | 5000     | 1.23   |
| 1002  | PEN         | Pen Blue  | 8000     | 1.25   |
| 1003  | PEN         | Pen Black | 2000     | 1.25   |
| 1004  | PEC         | Pencil 2B | 10000    | 0.48   |
| 1005  | PEC         | Pencil 2H | 8000     | 0.49   |
| 1006  | PEC         | Pencil HB | 0         | 99999.99 |
+-----+-----+-----+-----+-----+
6 rows in set (0.001 sec)

MariaDB [storedb]> DELETE FROM products WHERE id = 1006;
Query OK, 1 row affected (0.016 sec)

MariaDB [storedb]> |
```

Pembahasan : Program praktik 4 membahas perintah INSERT INTO untuk menambahkan data ke tabel products. Terdapat tiga cara insert yang didemonstrasikan, pertama memasukkan semua kolom dengan nilai lengkap termasuk id, kedua memasukkan beberapa baris sekaligus dengan id diisi NULL agar AUTO_INCREMENT yang mengisinya otomatis, ketiga memasukkan data hanya pada kolom tertentu saja dengan menyebutkan nama kolomnya secara

eksplisit. Perintah DELETE FROM products WHERE id = 1006 digunakan untuk menghapus satu baris data yang id-nya 1006.

Praktik 5

```
MariaDB [storedb]> SELECT name, price FROM products;
```

name	price
Pen Red	1.23
Pen Blue	1.25
Pen Black	1.25
Pencil 2B	0.48
Pencil 2H	0.49

```
5 rows in set (0.001 sec)
```



```
MariaDB [storedb]> SELECT * FROM products;
```

id	productCode	name	quantity	price
1001	PEN	Pen Red	5000	1.23
1002	PEN	Pen Blue	8000	1.25
1003	PEN	Pen Black	2000	1.25
1004	PEC	Pencil 2B	10000	0.48
1005	PEC	Pencil 2H	8000	0.49

```
5 rows in set (0.000 sec)
```



```
MariaDB [storedb]> SELECT 1+1;
```

1+1
2

```
1 row in set (0.000 sec)
```



```
MariaDB [storedb]> SELECT NOW();
```

NOW()
2026-05-06 10:41:25

```
1 row in set (0.000 sec)
```



```
MariaDB [storedb]> SELECT 1+1, NOW();
```

1+1	NOW()
2	2026-05-06 10:41:25

```
1 row in set (0.000 sec)
```



```
MariaDB [storedb]> SELECT name, price FROM products WHERE price < 1.0;
```

name	price
------	-------


```
Administrator: C:\WINDOWS\ × + ∨

MariaDB [storedb]> SELECT name, price FROM products WHERE price < 1.0;
+-----+-----+
| name      | price |
+-----+-----+
| Pencil 2B | 0.48  |
| Pencil 2H | 0.49  |
+-----+-----+
2 rows in set (0.015 sec)

MariaDB [storedb]> SELECT name, quantity FROM products WHERE quantity <= 2000;
+-----+-----+
| name      | quantity |
+-----+-----+
| Pen Black | 2000     |
+-----+-----+
1 row in set (0.000 sec)

MariaDB [storedb]> SELECT name, price FROM products WHERE productCode = 'PEN';
+-----+-----+
| name      | price |
+-----+-----+
| Pen Red   | 1.23  |
| Pen Blue  | 1.25  |
| Pen Black | 1.25  |
+-----+-----+
3 rows in set (0.000 sec)

MariaDB [storedb]> SELECT name, price FROM products WHERE name LIKE 'PENCIL%';
+-----+-----+
| name      | price |
+-----+-----+
| Pencil 2B | 0.48  |
| Pencil 2H | 0.49  |
+-----+-----+
2 rows in set (0.000 sec)

MariaDB [storedb]> SELECT name, price FROM products WHERE name LIKE 'P__ %';
+-----+-----+
| name      | price |
+-----+-----+
| Pen Red   | 1.23  |
| Pen Blue  | 1.25  |
| Pen Black | 1.25  |
+-----+-----+
3 rows in set (0.000 sec)

MariaDB [storedb]> SELECT * FROM products WHERE quantity >= 5000 AND name LIKE 'Pen %';
+-----+-----+-----+-----+-----+
| id  | productCode | name      | quantity | price |
+-----+-----+-----+-----+-----+
| 1001 | PEN         | Pen Red   | 5000     | 1.23  |
| 1002 | PEN         | Pen Blue  | 8000     | 1.25  |
+-----+-----+-----+-----+-----+
```

```
Administrator: C:\WINDOWS\ ...
2 rows in set (0.000 sec)

MariaDB [storedb]> SELECT * FROM products WHERE quantity >= 5000 AND price < 1.24 AND name LIKE 'Pen %';
+----+-----+-----+-----+-----+
| id | productCode | name | quantity | price |
+----+-----+-----+-----+
| 1001 | PEN | Pen Red | 5000 | 1.23 |
+----+-----+-----+-----+
1 row in set (0.001 sec)

MariaDB [storedb]> SELECT * FROM products WHERE NOT (quantity >= 5000 AND name LIKE 'Pen %');
+----+-----+-----+-----+-----+
| id | productCode | name | quantity | price |
+----+-----+-----+-----+
| 1003 | PEN | Pen Black | 2000 | 1.25 |
| 1004 | PEC | Pencil 2B | 10000 | 0.48 |
| 1005 | PEC | Pencil 2H | 8000 | 0.49 |
+----+-----+-----+-----+
3 rows in set (0.000 sec)

MariaDB [storedb]> SELECT * FROM products WHERE name IN ('Pen Red', 'Pen Black');
+----+-----+-----+-----+-----+
| id | productCode | name | quantity | price |
+----+-----+-----+-----+
| 1001 | PEN | Pen Red | 5000 | 1.23 |
| 1003 | PEN | Pen Black | 2000 | 1.25 |
+----+-----+-----+-----+
2 rows in set (0.001 sec)

MariaDB [storedb]> SELECT * FROM products WHERE (price BETWEEN 1.0 AND 2.0) AND (quantity BETWEEN 1000 AND 2000);
+----+-----+-----+-----+-----+
| id | productCode | name | quantity | price |
+----+-----+-----+-----+
| 1003 | PEN | Pen Black | 2000 | 1.25 |
+----+-----+-----+-----+
1 row in set (0.000 sec)

MariaDB [storedb]> SELECT * FROM products WHERE productCode IS NULL;
Empty set (0.000 sec)

MariaDB [storedb]> SELECT * FROM products WHERE name LIKE 'Pen %' ORDER BY price DESC;
+----+-----+-----+-----+-----+
| id | productCode | name | quantity | price |
+----+-----+-----+-----+
| 1002 | PEN | Pen Blue | 8000 | 1.25 |
| 1003 | PEN | Pen Black | 2000 | 1.25 |
| 1001 | PEN | Pen Red | 5000 | 1.23 |
+----+-----+-----+-----+
3 rows in set (0.000 sec)

MariaDB [storedb]> SELECT * FROM products WHERE name LIKE 'Pen %' ORDER BY price DESC, quantity;
+----+-----+-----+-----+-----+
| id | productCode | name | quantity | price |
+----+-----+-----+-----+
| 1002 | PEN | Pen Blue | 8000 | 1.25 |
| 1003 | PEN | Pen Black | 2000 | 1.25 |
| 1001 | PEN | Pen Red | 5000 | 1.23 |
+----+-----+-----+-----+
```

```
Administrator: C:\WINDOWS\
+ + +
MariaDB [storedb]> SELECT * FROM products WHERE name LIKE 'Pen %' ORDER BY price DESC, quantity;
+----+-----+-----+-----+-----+
| id | productCode | name | quantity | price |
+----+-----+-----+-----+-----+
| 1003 | PEN | Pen Black | 2000 | 1.25 |
| 1002 | PEN | Pen Blue | 8000 | 1.25 |
| 1001 | PEN | Pen Red | 5000 | 1.23 |
+----+-----+-----+-----+-----+
3 rows in set (0.000 sec)

MariaDB [storedb]> SELECT * FROM products ORDER BY RAND();
+----+-----+-----+-----+-----+
| id | productCode | name | quantity | price |
+----+-----+-----+-----+-----+
| 1004 | PEC | Pencil 2B | 10000 | 0.48 |
| 1003 | PEN | Pen Black | 2000 | 1.25 |
| 1001 | PEN | Pen Red | 5000 | 1.23 |
| 1002 | PEN | Pen Blue | 8000 | 1.25 |
| 1005 | PEC | Pencil 2H | 8000 | 0.49 |
+----+-----+-----+-----+-----+
5 rows in set (0.001 sec)

MariaDB [storedb]> SELECT * FROM products ORDER BY price LIMIT 2;
+----+-----+-----+-----+-----+
| id | productCode | name | quantity | price |
+----+-----+-----+-----+-----+
| 1004 | PEC | Pencil 2B | 10000 | 0.48 |
| 1005 | PEC | Pencil 2H | 8000 | 0.49 |
+----+-----+-----+-----+-----+
2 rows in set (0.000 sec)

MariaDB [storedb]> SELECT * FROM products ORDER BY price LIMIT 2, 1;
+----+-----+-----+-----+-----+
| id | productCode | name | quantity | price |
+----+-----+-----+-----+-----+
| 1001 | PEN | Pen Red | 5000 | 1.23 |
+----+-----+-----+-----+-----+
1 row in set (0.000 sec)

MariaDB [storedb]> SELECT id AS ID, productCode AS Code, name AS Description, price AS 'Unit Price' FROM products ORDER BY ID;
+----+-----+-----+-----+
| ID | Code | Description | Unit Price |
+----+-----+-----+-----+
| 1001 | PEN | Pen Red | 1.23 |
| 1002 | PEN | Pen Blue | 1.25 |
| 1003 | PEN | Pen Black | 1.25 |
| 1004 | PEC | Pencil 2B | 0.48 |
| 1005 | PEC | Pencil 2H | 0.49 |
+----+-----+-----+-----+
5 rows in set (0.000 sec)

MariaDB [storedb]> SELECT CONCAT(productCode, ' - ', name) AS 'Product Description', price FROM products;
+-----+-----+
| Product Description | price |
+-----+-----+
| 1001 - Pen Red | 1.23 |
| 1002 - Pen Blue | 1.25 |
| 1003 - Pen Black | 1.25 |
| 1004 - Pencil 2B | 0.48 |
| 1005 - Pencil 2H | 0.49 |
+-----+-----+
5 rows in set (0.000 sec)
```

```
Administrator: C:\WINDOWS\ × + v
MariaDB [storedb]> SELECT CONCAT(productCode, ' - ', name) AS 'Product Description', price FROM products;
+-----+-----+
| Product Description | price |
+-----+-----+
| PEN - Pen Red      | 1.23  |
| PEN - Pen Blue     | 1.25  |
| PEN - Pen Black     | 1.25  |
| PEC - Pencil 2B     | 0.48  |
| PEC - Pencil 2H     | 0.49  |
+-----+-----+
5 rows in set (0.001 sec)

MariaDB [storedb]> SELECT price FROM products;
+-----+
| price |
+-----+
| 1.23  |
| 1.25  |
| 1.25  |
| 0.48  |
| 0.49  |
+-----+
5 rows in set (0.000 sec)

MariaDB [storedb]> SELECT DISTINCT price AS 'Distinct Price' FROM products;
+-----+
| Distinct Price |
+-----+
| 1.23           |
| 1.25           |
| 0.48           |
| 0.49           |
+-----+
4 rows in set (0.000 sec)

MariaDB [storedb]> SELECT DISTINCT price, name FROM products;
+-----+-----+
| price | name |
+-----+-----+
| 1.23  | Pen Red |
| 1.25  | Pen Blue |
| 1.25  | Pen Black |
| 0.48  | Pencil 2B |
| 0.49  | Pencil 2H |
+-----+-----+
5 rows in set (0.001 sec)

MariaDB [storedb]> SELECT * FROM products ORDER BY productCode, id;
+-----+-----+-----+-----+-----+
| id   | productCode | name      | quantity | price |
+-----+-----+-----+-----+-----+
| 1004 | PEC         | Pencil 2B | 10000    | 0.48  |
| 1005 | PEC         | Pencil 2H | 8000     | 0.49  |
+-----+-----+-----+-----+-----+
```

```
Administrator: C:\WINDOWS\ ... + v
MariaDB [storedb]> SELECT * FROM products ORDER BY productCode, id;
+-----+-----+-----+-----+-----+
| id | productCode | name | quantity | price |
+-----+-----+-----+-----+-----+
| 1004 | PEC | Pencil 2B | 10000 | 0.48 |
| 1005 | PEC | Pencil 2H | 8000 | 0.49 |
| 1001 | PEN | Pen Red | 5000 | 1.23 |
| 1002 | PEN | Pen Blue | 8000 | 1.25 |
| 1003 | PEN | Pen Black | 2000 | 1.25 |
+-----+-----+-----+-----+-----+
5 rows in set (0.000 sec)

MariaDB [storedb]> SELECT * FROM products GROUP BY productCode;
+-----+-----+-----+-----+-----+
| id | productCode | name | quantity | price |
+-----+-----+-----+-----+-----+
| 1004 | PEC | Pencil 2B | 10000 | 0.48 |
| 1001 | PEN | Pen Red | 5000 | 1.23 |
+-----+-----+-----+-----+-----+
2 rows in set (0.001 sec)

MariaDB [storedb]> SELECT COUNT(*) AS 'Count' FROM products;
+-----+
| Count |
+-----+
| 5 |
+-----+
1 row in set (0.000 sec)

MariaDB [storedb]> SELECT productCode, COUNT(*) FROM products GROUP BY productCode;
+-----+-----+
| productCode | COUNT(*) |
+-----+-----+
| PEC | 2 |
| PEN | 3 |
+-----+-----+
2 rows in set (0.000 sec)

MariaDB [storedb]> SELECT productCode, COUNT(*) AS count FROM products GROUP BY productCode ORDER BY count DESC;
+-----+-----+
| productCode | count |
+-----+-----+
| PEN | 3 |
| PEC | 2 |
+-----+-----+
2 rows in set (0.001 sec)

MariaDB [storedb]> SELECT MAX(price), MIN(price), AVG(price), STD(price), SUM(quantity) FROM products;
+-----+-----+-----+-----+-----+
| MAX(price) | MIN(price) | AVG(price) | STD(price) | SUM(quantity) |
+-----+-----+-----+-----+-----+
| 1.25 | 0.48 | 0.940000 | 0.371591 | 33000 |
+-----+-----+-----+-----+-----+
2 rows in set (0.001 sec)
```

```
Administrator: C:\WINDOWS\ ... + v
+-----+-----+
| PEN | 3 |
| PEC | 2 |
+-----+-----+
2 rows in set (0.001 sec)

MariaDB [storedb]> SELECT MAX(price), MIN(price), AVG(price), STD(price), SUM(quantity) FROM products;
+-----+-----+-----+-----+-----+
| MAX(price) | MIN(price) | AVG(price) | STD(price) | SUM(quantity) |
+-----+-----+-----+-----+-----+
| 1.25 | 0.48 | 0.940000 | 0.371591 | 33000 |
+-----+-----+-----+-----+-----+
1 row in set (0.001 sec)

MariaDB [storedb]> SELECT productCode, MAX(price) AS 'Highest Price', MIN(price) AS 'Lowest Price' FROM products GROUP BY productCode;
+-----+-----+-----+
| productCode | Highest Price | Lowest Price |
+-----+-----+-----+
| PEC | 0.49 | 0.48 |
| PEN | 1.25 | 1.23 |
+-----+-----+-----+
2 rows in set (0.000 sec)

MariaDB [storedb]> SELECT productCode, MAX(price), MIN(price), CAST(AVG(price) AS DECIMAL(7,2)) AS 'Average', CAST(STD(price) AS DECIMAL(7,2)) AS 'Std Dev', SUM(quantity) FROM products GROUP BY productCode;
+-----+-----+-----+-----+-----+-----+
| productCode | MAX(price) | MIN(price) | Average | Std Dev | SUM(quantity) |
+-----+-----+-----+-----+-----+-----+
| PEC | 0.49 | 0.48 | 0.49 | 0.01 | 18000 |
| PEN | 1.25 | 1.23 | 1.24 | 0.01 | 15000 |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.001 sec)

MariaDB [storedb]> SELECT productCode AS 'Product Code', COUNT(*) AS 'Count', CAST(AVG(price) AS DECIMAL(7,2)) AS 'Average' FROM products GROUP BY productCode HAVING Count >=3;
+-----+-----+-----+
| Product Code | Count | Average |
+-----+-----+-----+
| PEN | 3 | 1.24 |
+-----+-----+-----+
1 row in set (0.001 sec)

MariaDB [storedb]> SELECT productCode, MAX(price), MIN(price), CAST(AVG(price) AS DECIMAL(7,2)) AS 'Average', SUM(quantity) FROM products GROUP BY productCode WITH ROLLUP;
+-----+-----+-----+-----+-----+
| productCode | MAX(price) | MIN(price) | Average | SUM(quantity) |
+-----+-----+-----+-----+-----+
| PEC | 0.49 | 0.48 | 0.49 | 18000 |
| PEN | 1.25 | 1.23 | 1.24 | 15000 |
| NULL | 1.25 | 0.48 | 0.94 | 33000 |
+-----+-----+-----+-----+-----+
3 rows in set (0.001 sec)

MariaDB [storedb]> |
```

Pembahasan : Program praktik 5 membahas berbagai variasi perintah SELECT untuk mengambil data dari tabel products. Perintah SELECT * mengambil semua kolom sedangkan SELECT name, price hanya mengambil kolom tertentu. Klausa WHERE digunakan untuk memfilter data dengan berbagai operator seperti <, <=, =, LIKE untuk pattern matching, AND dan OR untuk logika ganda, IN untuk mencocokkan beberapa nilai sekaligus, dan BETWEEN untuk range nilai. Klausa ORDER BY mengurutkan hasil, LIMIT membatasi jumlah baris yang ditampilkan, AS memberi alias nama kolom, DISTINCT menghilangkan duplikat, dan GROUP BY mengelompokkan data yang dikombinasikan dengan fungsi agregat seperti COUNT, MAX, MIN, AVG, dan SUM

Praktik 6

```
MariaDB [storedb]> UPDATE products SET price = price * 1.1;
Query OK, 5 rows affected, 5 warnings (0.003 sec)
Rows matched: 5 Changed: 5 Warnings: 5

MariaDB [storedb]> SELECT * FROM products;
+----+-----+-----+-----+-----+
| id | productCode | name | quantity | price |
+----+-----+-----+-----+-----+
| 1001 | PEN | Pen Red | 5000 | 1.35 |
| 1002 | PEN | Pen Blue | 8000 | 1.38 |
| 1003 | PEN | Pen Black | 2000 | 1.38 |
| 1004 | PEC | Pencil 2B | 10000 | 0.53 |
| 1005 | PEC | Pencil 2H | 8000 | 0.54 |
+----+-----+-----+-----+-----+
5 rows in set (0.000 sec)

MariaDB [storedb]> UPDATE products SET quantity = quantity - 100 WHERE name = 'Pen Red';
Query OK, 1 row affected (0.002 sec)
Rows matched: 1 Changed: 1 Warnings: 0

MariaDB [storedb]> SELECT * FROM products WHERE name = 'Pen Red';
+----+-----+-----+-----+-----+
| id | productCode | name | quantity | price |
+----+-----+-----+-----+-----+
| 1001 | PEN | Pen Red | 4900 | 1.35 |
+----+-----+-----+-----+-----+
1 row in set (0.000 sec)

MariaDB [storedb]> UPDATE products SET quantity = quantity + 50, price = 1.23 WHERE name = 'Pen Red';
Query OK, 1 row affected (0.002 sec)
Rows matched: 1 Changed: 1 Warnings: 0

MariaDB [storedb]> SELECT * FROM products WHERE name = 'Pen Red';
+----+-----+-----+-----+-----+
| id | productCode | name | quantity | price |
+----+-----+-----+-----+-----+
| 1001 | PEN | Pen Red | 4950 | 1.23 |
+----+-----+-----+-----+-----+
1 row in set (0.001 sec)

MariaDB [storedb]> |
```

Pembahasan : Program praktik 6 membahas perintah UPDATE untuk mengubah data yang sudah ada di tabel products. Perintah UPDATE products SET price = price * 1.1 mengubah seluruh baris pada kolom price menjadi 110% dari nilai sebelumnya tanpa klausa WHERE sehingga semua baris ikut berubah. Perintah UPDATE products SET quantity = quantity - 100 WHERE name = 'Pen Red' hanya mengubah baris yang kolom name bernilai 'Pen Red', dan UPDATE dengan dua kolom sekaligus menggunakan tanda koma untuk memisahkan setiap kolom yang diubah.

Praktik 7

```
MariaDB [storedb]>
MariaDB [storedb]> DELETE FROM products WHERE name LIKE 'Pencil%';
Query OK, 2 rows affected (0.004 sec)

MariaDB [storedb]> SELECT * FROM products;
+-----+-----+-----+-----+-----+
| id | productCode | name | quantity | price |
+-----+-----+-----+-----+-----+
| 1001 | PEN | Pen Red | 4950 | 1.23 |
| 1002 | PEN | Pen Blue | 8000 | 1.38 |
| 1003 | PEN | Pen Black | 2000 | 1.38 |
+-----+-----+-----+-----+-----+
3 rows in set (0.016 sec)

MariaDB [storedb]> DELETE FROM products;
Query OK, 3 rows affected (0.002 sec)

MariaDB [storedb]> SELECT * FROM products;
Empty set (0.001 sec)

MariaDB [storedb]> |
```

Pembahasan : Program praktik 7 membahas perintah DELETE FROM untuk menghapus data dari tabel products. Perintah DELETE FROM products WHERE name LIKE 'Pencil%' menghapus semua baris yang nilai kolom name diawali kata "Pencil" menggunakan operator LIKE dengan wildcard %. Perintah DELETE FROM products tanpa klausa WHERE menghapus seluruh isi tabel sekaligus sehingga tabel menjadi kosong namun struktur tabelnya tetap ada.

Praktik 8

```
MariaDB [storedb]>
MariaDB [storedb]>
MariaDB [storedb]> LOAD DATA LOCAL INFILE 'd:/myProject/products_in.csv' INTO TABLE products COLUMNS TERMINATED BY ',' LINES TERMINATED BY '\r\n';
Query OK, 0 rows affected, 5 warnings (0.019 sec)
Records: 5 Deleted: 0 Skipped: 5 Warnings: 5

MariaDB [storedb]> SELECT * FROM products;
+-----+-----+-----+-----+-----+
| id | productCode | name | quantity | price |
+-----+-----+-----+-----+-----+
| 1001 | PEN | Pen Red | 5000 | 1.23 |
| 1007 | PEN | Pen Blue | 8000 | 1.25 |
| 1008 | PEN | Pen Black | 2000 | 1.25 |
| 1009 | PEC | Pencil 2B | 10000 | 0.48 |
| 1010 | PEC | Pencil 2H | 8000 | 0.49 |
+-----+-----+-----+-----+-----+
5 rows in set (0.008 sec)

MariaDB [storedb]> SELECT * FROM products INTO OUTFILE 'd:/myProject/products_out.csv' COLUMNS TERMINATED BY ',' LINES TERMINATED BY '\r\n';
ERROR 1086 (HY000): File 'd:/myProject/products_out.csv' already exists
MariaDB [storedb]> SELECT * FROM products INTO OUTFILE 'd:/myProject/products_out.csv' COLUMNS TERMINATED BY ',' LINES TERMINATED BY '\r\n';
Query OK, 5 rows affected, 1 warning (0.002 sec)
```

Pembahasan : Program praktik 8 membahas cara import dan export data menggunakan file CSV. Perintah LOAD DATA LOCAL INFILE digunakan untuk mengimpor data dari file CSV ke dalam tabel products dengan COLUMNS TERMINATED BY ',' menandakan pemisah antar kolom adalah koma dan LINES TERMINATED BY '\r\n' menandakan pemisah antar baris. Perintah SELECT * FROM products INTO OUTFILE digunakan untuk mengekspor seluruh isi tabel ke file CSV dengan format yang sama. File yang akan diimport harus sudah ada terlebih dahulu di path yang disebutkan, dan file export tidak boleh sudah ada sebelumnya karena MariaDB tidak akan menimpa file yang sudah ada.

Praktik 9

```
MariaDB [storedb]> USE storedb;
Database changed
MariaDB [storedb]> DROP TABLE IF EXISTS suppliers;
Query OK, 0 rows affected, 1 warning (0.001 sec)

MariaDB [storedb]> CREATE TABLE suppliers (
  -> supplierID INT UNSIGNED NOT NULL AUTO_INCREMENT,
  -> name VARCHAR(30) NOT NULL DEFAULT '',
  -> phone CHAR(8) NOT NULL DEFAULT '',
  -> PRIMARY KEY (supplierID)
  -> );
Query OK, 0 rows affected (0.041 sec)

MariaDB [storedb]> DESCRIBE suppliers;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| supplierID | int(10) unsigned | NO | PRI | NULL | auto_increment |
| name | varchar(30) | NO | | | |
| phone | char(8) | NO | | | |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.021 sec)

MariaDB [storedb]> INSERT INTO suppliers VALUES (501, 'ABC Traders', '88881111'), (502, 'XYZ Company', '88882222'), (503, 'QQ Corp', '88883333');
Query OK, 3 rows affected (0.003 sec)
Records: 3 Duplicates: 0 Warnings: 0

MariaDB [storedb]> SELECT * FROM suppliers;
+-----+-----+-----+
| supplierID | name | phone |
+-----+-----+-----+
| 501 | ABC Traders | 88881111 |
| 502 | XYZ Company | 88882222 |
| 503 | QQ Corp | 88883333 |
+-----+-----+-----+
3 rows in set (0.002 sec)

MariaDB [storedb]> ALTER TABLE products ADD COLUMN supplierID INT UNSIGNED NOT NULL;
Query OK, 0 rows affected (0.011 sec)
Records: 0 Duplicates: 0 Warnings: 0

MariaDB [storedb]> DESCRIBE products;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| id | int(10) unsigned | NO | PRI | NULL | auto_increment |
| productCode | char(3) | NO | | | |
| name | varchar(30) | NO | | | |
| quantity | int(10) unsigned | NO | | 0 | |
| price | decimal(7,2) | NO | | 99999.99 | |
| supplierID | int(10) unsigned | NO | | NULL | |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.008 sec)

MariaDB [storedb]> UPDATE products SET supplierID = 501;
Query OK, 5 rows affected (0.007 sec)
Rows matched: 5 Changed: 5 Warnings: 0

MariaDB [storedb]> ALTER TABLE products ADD FOREIGN KEY (supplierID) REFERENCES suppliers (supplierID);
Query OK, 5 rows affected (0.067 sec)
Records: 5 Duplicates: 0 Warnings: 0

MariaDB [storedb]> DESCRIBE products;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| id | int(10) unsigned | NO | PRI | NULL | auto_increment |
| productCode | char(3) | NO | | | |
| name | varchar(30) | NO | | | |
| quantity | int(10) unsigned | NO | | 0 | |
| price | decimal(7,2) | NO | | 99999.99 | |
| supplierID | int(10) unsigned | NO | MUL | NULL | |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.008 sec)

MariaDB [storedb]> UPDATE products SET supplierID = 502 WHERE id = 2004;
Query OK, 0 rows affected (0.001 sec)
Rows matched: 0 Changed: 0 Warnings: 0

MariaDB [storedb]> SELECT * FROM products;
+-----+-----+-----+-----+-----+-----+
| id | productCode | name | quantity | price | supplierID |
+-----+-----+-----+-----+-----+-----+
| 1001 | PEN | Pen Red | 5000 | 1.23 | 501 |
| 1007 | PEN | Pen Blue | 8000 | 1.25 | 501 |
| 1008 | PEN | Pen Black | 2000 | 1.25 | 501 |
| 1009 | PEC | Pencil 2B | 10000 | 0.48 | 501 |
| 1010 | PEC | Pencil 2H | 8000 | 0.49 | 501 |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.008 sec)

MariaDB [storedb]> SELECT products.name, price, suppliers.name FROM products JOIN suppliers ON products.supplierID = suppliers.supplierID WHERE price < 0.6;
+-----+-----+-----+
| name | price | name |
+-----+-----+-----+
| Pencil 2B | 0.48 | ABC Traders |
| Pencil 2H | 0.49 | ABC Traders |
+-----+-----+-----+
2 rows in set (0.001 sec)

MariaDB [storedb]> SELECT products.name, price, suppliers.name FROM products, suppliers WHERE products.supplierID = suppliers.supplierID AND price < 0.6;
+-----+-----+-----+
| name | price | name |
+-----+-----+-----+
| Pencil 2B | 0.48 | ABC Traders |
| Pencil 2H | 0.49 | ABC Traders |
+-----+-----+-----+
```

Pembahasan : Program praktik 9 membahas relasi antar tabel menggunakan foreign key. Tabel suppliers dibuat dengan kolom supplierID sebagai primary key, name, dan phone, kemudian diisi tiga data supplier. Tabel products ditambah kolom supplierID menggunakan

ALTER TABLE products ADD COLUMN supplierID lalu dihubungkan ke tabel suppliers menggunakan ALTER TABLE products ADD FOREIGN KEY (supplierID) REFERENCES suppliers (supplierID) sehingga nilai supplierID di tabel products harus merujuk ke supplierID yang ada di tabel suppliers. Perintah SELECT dengan JOIN digunakan untuk menampilkan data dari kedua tabel sekaligus dengan mencocokkan kolom supplierID yang sama, dan penggunaan alias AS p dan AS s membuat penulisan query lebih singkat.

C. PEMBAHASAN TUGAS

D. KESIMPULAN

Dari praktikum ini dapat disimpulkan bahwa SQL menyediakan perintah lengkap untuk operasi CRUD yaitu INSERT untuk menambah data, SELECT untuk membaca data, UPDATE untuk mengubah data, dan DELETE untuk menghapus data. Selain itu SQL juga menyediakan perintah untuk manajemen database dan tabel seperti CREATE, DROP, dan ALTER. Penggunaan klausa WHERE sangat penting untuk membatasi baris yang terpengaruh oleh perintah UPDATE dan DELETE agar tidak mengubah atau menghapus seluruh data secara tidak sengaja. Relasi antar tabel menggunakan foreign key memungkinkan data dari dua tabel berbeda digabungkan menggunakan perintah JOIN.

E. LAMPIRAN